

CSE 3110 - Database Systems Laboratory

PROJECT PROPOSAL

ROOMORA

Hotel Booking & Comparison Web Application

Submitted By

Name: Md. Sadikul Islam Siyam

Roll No: 2207031

Department: Computer Science
& Engineering

1. Architectural Overview & System Design

Roomora is a high-performance hotel booking and comparison platform designed using the **Model-View-Controller (MVC)** architectural pattern native to Laravel 12. The application cleanly decouples business logic, user interfaces, and HTTP request handling to ensure maintainability, testability, and clean code division across engineering iterations.

To prevent controller bloat, complex processes—such as multi-room availability validation patterns and dynamic stay pricing calculation engines—are isolated into dedicated Service layers. The frontend utilizes an AJAX-driven interface pattern to ensure real-time search queries, live contextual lookup filters, and favorite/wishlist state toggles occur instantly without page lifecycles resetting.

2. Normalized Database Design (3NF)

The database schema has been engineered to strict **Third Normal Form (3NF)** standards to systematically remove structural anomalies, enforce referential data chains, and ensure optimal execution indexing across deep information searches.

Relational Database Schema Summary

- **users:** id (PK), name, email (Unique, Index), phone, password, role (Admin/Customer, Index), avatar, softDeletes, timestamps.
- **hotels:** id (PK), name (Index), city (Index), address, description, star_rating, image, softDeletes, timestamps.
- **rooms:** id (PK), hotel_id (FK, Index), room_type, price (Index), capacity, facilities (JSON), quantity, softDeletes, timestamps.
- **bookings:** id (PK), user_id (FK, Index), room_id (FK, Index), check_in, check_out, guests, total_price, status (Index), softDeletes, timestamps.
- **reviews:** id (PK), user_id (FK), hotel_id (FK), rating, comment, timestamps.
- **comparisons:** id (PK), user_id (FK), hotel_id (FK), timestamps. (Unique compound key across user and hotel mappings).
- **favorites:** id (PK), user_id (FK), hotel_id (FK), timestamps. (Unique compound key across user and hotel mappings).

3. Technology Stack & Security Paradigm

The Development Stack

- **Backend Framework:** Laravel 12 (PHP 8.2) utilizing robust Eloquent ORM abstraction layers.
- **Database Tier:** MySQL 8.0 executing complete transactional ACID boundaries.
- **Frontend Architecture:** Blade Template Engine and Bootstrap 5 for liquid layout viewport scaling.
- **Reporting Elements:** Chart.js canvas graph loops and barryvdh/laravel-dompdf for secure invoice generation.

Comprehensive Security & Admin Privileges

- **Cryptographic Salting:** Passwords map exclusively to secure `bcrypt Argon2id` hash patterns.
- **XSS Mitigation:** Explicit rendering protections utilizing Blade double escaping syntax `{{ }}` backed by `strip_tags()` input sanitization templates.
- **SQL Injection Shield:** Isolation of active request parameters by enforcing parameterization across all ORM commands.
- **Role-Based Access Control (RBAC):** Isolation of general user areas from privileged administrative paths using dedicated authentication middleware and custom Laravel policies.

4. Entity Relationship Diagram (ERD) & Schema Visualizer

The relational map below visually breaks down the underlying 3NF entity tables. Primary Keys (**PK**) and Foreign Keys (**FK**) dictate relational cascading boundaries.



ER Diagram Structural Logic

- **User Interactions:** A user profile coordinates entries across transactional mappings like bookings, reviews, and favorites. If an account is soft-deleted, information histories can be recovered.
- **Hotel Inventory Boundaries:** Removing an entry in hotels triggers a cascading drop across dependent room elements.

- **Booking Ledger Rules:** To protect financial histories for audits, rooms implement `ON DELETE RESTRICT` constraints within active reservation schemas. Overlap validations assess existing date ranges before confirming check-out blocks.

5. Core Functional Modules (Feature Matrix)

User & Engagement Modules

- **Profile Management:** User registration, secure login validation loops, avatar uploads, and persistent reservation tracking grids.
- **Wishlist & Bookmarks:** A single-click asynchronous engine to add targeted hotel configurations to a user's personal profile view.
- **Verified Review Engine:** Enforces check boundaries via Eloquent to limit review submissions exclusively to users with a verified `checked_out` status, drawing dynamic breakdown charts.

Discovery & Comparison Modules

- **AJAX Search & Filter Architecture:** Asynchronous live lookup bar providing real-time property suggestions as the user types, with filter parameters covering prices, star-ratings, and amenities.
- **Side-by-Side Comparison Matrix:** Structured layout contrasting up to 4 hotels side-by-side based on exact price points, amenities, checkout timing rules, and room types.

Admin Module & Operational Management Tools

- **Role-Based Access Control (RBAC):** Implements strict custom middleware and policies checking user roles (Customer vs. Admin) before providing workspace permissions.
- **Live Telemetry Dashboard:** Administrative command panel capturing total registered users, active hotels, aggregate bookings, and monthly financial revenues visualized via custom Chart.js lines and bars.
- **Full System CRUD & Configuration Control:** Dedicated control dashboards providing administrators complete CRUD capabilities across users, hotels, rooms, and bookings.
- **Booking Lifecycle State Machine:** Administrative workflow handlers driving sequential reservation stages (pending → confirmed → checked_in → checked_out).
- **Financial Auditing & Data Exports:** Complete dynamic transactional report generation tool providing clean CSV data spreadsheet extraction for external bookkeeping reviews.

Transactional & Admin Modules

- **Booking Engine:** Real-time room quantity queries, live availability confirmations, dynamic pricing calculations, automated email updates, and formal cancellation loops.
- **Invoice Compilation Engine:** Converts confirmed reservation information into downloadable, secure PDF bills via laravel-dompdf.
- **Administrative Dashboard:** Unified panel mapping live business parameters (Total Users, Registered Hotels, Total Bookings, and Monthly Revenues) using dynamic Chart.js canvases.
- **Data Operations & Audits:** Comprehensive CRUD control layouts over core tables along with structural CSV data exports for transaction history records.